

34. Consider a file system where the records of the file are stored one after another both physically and logically. A record of the file can only be accessed by reading all the previous records. Design a C program to simulate the file allocation strategy.

AIM :

Consider a file system where the records of the file are stored one after another both physically and logically. A record of the file can only be accessed by reading all the previous records. Design a C program to simulate the file allocation strategy.

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
struct Record
```

```
{
```

```
    int recordNumber;
```

```
    char data[256];
```

```
};
```

```
int main()
```

```
{
```

```
    FILE *file;
```

```
    struct Record record;
```

```
    int recordNumber;
```

```
    /* Open or create file in write mode */
```

```
    file = fopen("sequential_file.txt", "wb");
```

```
    if (file == NULL)
```

```

{
    printf("Error opening the file.\n");
    return 1;
}

/* Write records sequentially */
printf("Enter records (Enter '0' as record number to exit):\n");

while (1)
{
    printf("Record Number: ");
    scanf("%d", &record.recordNumber);

    if (record.recordNumber == 0)
    {
        break;
    }

    printf("Data: ");
    scanf(" %[^\n]", record.data);

    fwrite(&record, sizeof(struct Record), 1, file);
}

fclose(file);

/* Reopen file in read mode */

```

```
file = fopen("sequential_file.txt", "rb");
```

```
if (file == NULL)
```

```
{
```

```
    printf("Error opening the file.\n");
```

```
    return 1;
```

```
}
```

```
/* Read a specific record */
```

```
while (1)
```

```
{
```

```
    printf("\nEnter the record number to read (0 to exit): ");
```

```
    scanf("%d", &recordNumber);
```

```
    if (recordNumber == 0)
```

```
    {
```

```
        break;
```

```
    }
```

```
    rewind(file);
```

```
while (fread(&record, sizeof(struct Record), 1, file))
```

```
{
```

```
    if (record.recordNumber == recordNumber)
```

```
    {
```

```
        printf("Record Number: %d\n", record.recordNumber);
```

```
        printf("Data: %s\n", record.data);
```

```
        break;
    }
}

if (feof(file))
{
    printf("Record not found.\n");
}
}

fclose(file);

return 0;
}
```

OUTPUT:

```
Enter records (Enter '0' as record number to exit):
Record Number: 101
Data: Alice
Record Number: 102
Data: Bob
Record Number: 103
Data: Charlie
Record Number: 0

Enter the record number to read (0 to exit): 102
Record Number: 102
Data: Bob
```